1. Show how to check the current runlevel on your system.

```
[root@localhost ~]# runlevel
N 5
[root@localhost ~]#
```

2. Use the tree or ls -R command to list the structure under /etc.

```
[root@localhost /]# ls -R /etc
/etc:
abrt                dbus-1                  gcrypt              issue                   modprobe.d          polkit-1            rsyslog.d           sysctl.d
adjtime             dconf                   gdbinit             issue.net               modules-load.d      popt.d              rwtab               systemd
aliases             default                 gdbinit.d           java                    motd                postfix             rwtab.d             system-release
aliases.db          depmod.d                gdm                 jvm                     mtab                ppp                 samba               system-release-cpe
alsa                dhcp                    geoclue             jvm-commmon             mtools.conf         prelink.conf.d      sane.d              tcsd.conf
alternatives        DIR_COLORS              GeoIP.conf          kdump.conf              multipath           printcap            sasl2               terminfo
anacrontab          DIR_COLORS.256color     GeoIP.conf.default  kernel                  my.cnf              profile             scl                 tmpfiles.d
asound.conf         DIR_COLORS.lightbgcolor ghostscript         krb5.conf               my.cnf.d            profile.d           securetty           trusted-key.key
at.deny             dleyna-server-service.conf gnome-vfs-2.0    krb5.conf.d             nanorc              protocols           security            tuned
audisp              dnsmasq.conf            gnupg               ksmtuned.conf           netconfig           pulse               selinux             udev
audit               dnsmasq.d               GREP_COLORS         ld.so.cache             NetworkManager      purple              services            udisks2
avahi               docker                  groff               ld.so.conf              networks            python              sestatus.conf       unbound
bash_completion.d   dracut.conf             group               ld.so.conf.d            nfs.conf            qemu-ga             setroubleshoot      updatedb.conf
bashrc              dracut.conf.d           group-              libaudit.conf           nfsmount.conf       qemu-kvm            setuptool.d         UPower
binfmt.d            drirc                   grub2.cfg           libblockdev             nsswitch.conf       radvd.conf          sgml                usb_modeswitch.conf
bluetooth           e2fsck.conf             grub.d              libibverbs.d            nsswitch.conf.bak   ras                 shadow              vconsole.conf
bonobo-activation   enscript.cfg            gshadow             libnl                   ntp                 rc0.d               shadow-             vimrc
brltty              environment             gshadow-            libreport               numad.conf          rc1.d               shells              virc
brltty.conf         ethertypes              gss                 libuser.conf            oddjob              rc2.d               skel                vmware-tools
chkconfig.d         exports                 gssproxy            libvirt                 oddjobd.conf        rc3.d               smartmontools       wgetrc
chrony.conf         exports.d               host.conf           locale.conf             oddjobd.conf.d      rc4.d               sos.conf            wpa_supplicant
chrony.keys         favicon.png             hostname            localtime               openldap            rc5.d               sound               wvdial.conf
cifs-utils          fcoe                    hosts               login.defs              opt                 rc6.d               speech-dispatcher   X11
cron.d              festival                hosts.allow         logrotate.conf          os-release          rc.d                ssh                 xdg
cron.daily          file02                  hosts.deny          logrotate.d             PackageKit          rc.local            ssl                 xinetd.d
cron.deny           file123                 hp                  lsm                     pam.d               rdma                statetab            xml
cron.hourly         filesystems             idmapd.conf         lvm                     passwd              redhat-release      statetab.d          yum
cron.monthly        firefox                 init.d              machine-id              passwd-             request-key.conf    subgid              yum.conf
crontab             firewalld               inittab             magic                   pbm2ppa.conf        request-key.d       subuid              yum.repos.d
cron.weekly         flatpak                 inputrc             mail.rc                 pinforc             resolv.conf         sudo.conf
crypttab            fonts                   iproute2            makedumpfile.conf.sample pkcs11             rhsm                sudoers
csh.cshrc           fprintd.conf            ipsec.conf          man_db.conf             pki                 rpc                 sudoers.d
csh.login           fstab                   ipsec.d             maven                   plymouth            rpm                 sudo-ldap.conf
cups                fuse.conf               ipsec.secrets       mcelog                  pm                  rsyncd.conf         sysconfig
cupshelpers         gconf                   iscsi               mke2fs.conf             pnm2ppa.conf        rsyslog.conf        sysctl.conf
```

3. Create a file, copy it to another location, then delete both.

```
[root@localhost sathish]# cd /one
[root@localhost one]# touch laddu
[root@localhost one]# ll
total 0
-rw-r--r--. 1 root root 0 May 17 15:29 laddu
[root@localhost one]# cp /one/laddu /home/sathish
[root@localhost one]# cd /home/sathish
[root@localhost sathish]# ll
total 0
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Desktop
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Documents
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Downloads
-rw-rw-r--. 1 sathish sathish 0 May 11 15:40 file11
-rw-r--r--. 1 root    root    0 May 17 15:29 laddu
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Music
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Pictures
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Public
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Templates
-rw-rw-r--. 1 sathish sathish 0 May 15 10:33 test123
drwxr-xr-x. 2 sathish sathish 6 May 10 19:49 Videos
[root@localhost sathish]# rm -rf /one/laddu
[root@localhost sathish]# rm -rf /home/sathish/laddu
[root@localhost sathish]#
```

4. Create a new file using touch, edit it with vi, and display its contents using cat.

```
[root@localhost sathish]# touch file0
[root@localhost sathish]# vi file0
[root@localhost sathish]# cat file0
hii
file0
[root@localhost sathish]#
```

5. Create a new partition using fdisk

```
[root@localhost ~]#
[root@localhost ~]# echo " - - - " >>/sys/class/scsi_host/host0/scan
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda             8:0    0   20G  0 disk
├sda1           8:1    0  400M  0 part /boot
└sda2           8:2    0 19.6G  0 part
  ├rhel-root 253:0    0 10.3G  0 lvm  /
  ├rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├rhel-home 253:2    0  956M  0 lvm  /home
  └rhel-var  253:3    0  6.5G  0 lvm  /var
sdb             8:16   0    2G  0 disk
└sdb2           8:18   0  400M  0 part /Nihal
sdc             8:32   0    1G  0 disk /sathii
sdd             8:48   0    2G  0 disk
├sdd1           8:49   0  200M  0 part /sri
├sdd2           8:50   0  400M  0 part /ram
└sdd3           8:51   0  600M  0 part /satya
sde             8:64   0    2G  0 disk
sr0            11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost ~]# fdisk /dev/sde
Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x16f34bd2.

Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p):
Using default response p
Partition number (1-4, default 1):
First sector (2048-4194303, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-4194303, default 4194303): +300M
Partition 1 of type Linux and of size 300 MiB is set

Command (m for help): wq
The partition table has been altered!

Calling ioctl() to re-read partition table.
Syncing disks.
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda             8:0    0   20G  0 disk
├sda1           8:1    0  400M  0 part /boot
└sda2           8:2    0 19.6G  0 part
  ├rhel-root 253:0    0 10.3G  0 lvm  /
  ├rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├rhel-home 253:2    0  956M  0 lvm  /home
  └rhel-var  253:3    0  6.5G  0 lvm  /var
sdb             8:16   0    2G  0 disk
└sdb2           8:18   0  400M  0 part /Nihal
sdc             8:32   0    1G  0 disk /sathii
sdd             8:48   0    2G  0 disk
├sdd1           8:49   0  200M  0 part /sri
├sdd2           8:50   0  400M  0 part /ram
└sdd3           8:51   0  600M  0 part /satya
sde             8:64   0    2G  0 disk
```

```
sr0             11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost ~]# mkdir /sdefile
[root@localhost ~]# mkfs.ext4 /dev/sde1
mke2fs 1.42.9 (28-Dec-2013)
Filesystem label=
OS type: Linux
Block size=1024 (log=0)
Fragment size=1024 (log=0)
Stride=0 blocks, Stripe width=0 blocks
76912 inodes, 307200 blocks
15360 blocks (5.00%) reserved for the super user
First data block=1
Maximum filesystem blocks=33947648
38 block groups
8192 blocks per group, 8192 fragments per group
2024 inodes per group
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]# blkid
/dev/mapper/rhel-var: UUID="978e27f0-7d57-4dc0-bb2d-a1d9346983bf" TYPE="xfs"
/dev/sda2: UUID="tP2dJZ-nC0k-UUg2-BwFA-gPfP-nnv6-0Cli1h" TYPE="LVM2_member"
/dev/sda1: UUID="ad4764ff-05af-4f49-ba8a-507a984872cb" TYPE="xfs"
/dev/sdb2: UUID="72921ac2-7642-4a64-8354-6907083f54f2" TYPE="xfs"
/dev/sdc: UUID="24b39253-0801-427c-b9fc-519651afc93e" TYPE="ext4"
/dev/sdd1: UUID="6c49d1f5-657c-47a5-9fe5-6229c53ff7ea" TYPE="ext4"
/dev/sdd2: UUID="012726bf-7604-451a-9631-2d33a4e9c7ea" TYPE="ext4"
/dev/sdd3: UUID="2ea6b801-346d-4d67-a990-76b2758d744b" TYPE="xfs"
/dev/sr0: UUID="2018-03-22-19-04-59-00" LABEL="RHEL-7.5 Server.x86_64" TYPE="iso9660" PTTYPE="dos"
/dev/mapper/rhel-root: UUID="d64ec991-efea-4da4-b418-aa0a07d3eaaf" TYPE="xfs"
/dev/mapper/rhel-swap: UUID="bbd1e301-23c2-4878-ba3c-f5c14fee2401" TYPE="swap"
/dev/mapper/rhel-home: UUID="a6d58447-ee81-40fc-89a4-a83bb3dc5f35" TYPE="xfs"
/dev/sde1: UUID="8e087049-9f9e-4afe-9468-e054a9d9c008" TYPE="ext4"
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# mount /sdefile
[root@localhost ~]# lsblk |grep -i sde
sde             8:64   0    2G  0 disk
└sde1           8:65   0  300M  0 part /sdefile
[root@localhost ~]#
```

6. Create three primary and one extended partition, then create a logical volume.

```
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda             8:0    0   20G  0 disk
├─sda1          8:1    0  400M  0 part /boot
└─sda2          8:2    0 19.6G  0 part
  ├─rhel-root 253:0    0 10.3G  0 lvm  /
  ├─rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├─rhel-home 253:2    0  956M  0 lvm  /home
  └─rhel-var  253:3    0  6.5G  0 lvm  /var
sdb             8:16   0    2G  0 disk
└─sdb2          8:18   0  400M  0 part /Nihal
sdc             8:32   0    1G  0 disk /sathii
sdd             8:48   0    3G  0 disk
sr0            11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost ~]# fdisk /dev/sdd
Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x8b0c4562.

Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p):
Using default response p
Partition number (1-4, default 1):
First sector (2048-6291455, default 2048):
Using default value 2048
Last sector, +sectors or +size{K,M,G} (2048-6291455, default 6291455): +100M
Partition 1 of type Linux and of size 100 MiB is set

Command (m for help): n
Partition type:
   p   primary (1 primary, 0 extended, 3 free)
   e   extended
Select (default p):
Using default response p
Partition number (2-4, default 2):
First sector (206848-6291455, default 206848):
Using default value 206848
Last sector, +sectors or +size{K,M,G} (206848-6291455, default 6291455): +100M
Partition 2 of type Linux and of size 100 MiB is set

Command (m for help): n
Partition type:
   p   primary (2 primary, 0 extended, 2 free)
   e   extended
Select (default p):
Using default response p
Partition number (3,4, default 3):
First sector (411648-6291455, default 411648):
Using default value 411648
Last sector, +sectors or +size{K,M,G} (411648-6291455, default 6291455): +100M
Partition 3 of type Linux and of size 100 MiB is set

Command (m for help): p

Disk /dev/sdd: 3221 MB, 3221225472 bytes, 6291456 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
Disk identifier: 0x8b0c4562

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1            2048      206847      102400   83  Linux
/dev/sdd2          206848      411647      102400   83  Linux
/dev/sdd3          411648      616447      102400   83  Linux

Command (m for help): n
Partition type:
   p   primary (3 primary, 0 extended, 1 free)
   e   extended
Select (default e):
Using default response e
Selected partition 4
First sector (616448-6291455, default 616448):
Using default value 616448
Last sector, +sectors or +size{K,M,G} (616448-6291455, default 6291455): +150M
Partition 4 of type Extended and of size 150 MiB is set

Command (m for help): p

Disk /dev/sdd: 3221 MB, 3221225472 bytes, 6291456 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x8b0c4562

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1            2048      206847      102400   83  Linux
/dev/sdd2          206848      411647      102400   83  Linux
/dev/sdd3          411648      616447      102400   83  Linux
/dev/sdd4          616448      923647      153600    5  Extended

Command (m for help): n
All primary partitions are in use
Adding logical partition 5
First sector (618496-923647, default 618496):
Using default value 618496
Last sector, +sectors or +size{K,M,G} (618496-923647, default 923647):
Using default value 923647
Partition 5 of type Linux and of size 149 MiB is set

Command (m for help): p

Disk /dev/sdd: 3221 MB, 3221225472 bytes, 6291456 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x8b0c4562

   Device Boot      Start         End      Blocks   Id  System
/dev/sdd1            2048      206847      102400   83  Linux
/dev/sdd2          206848      411647      102400   83  Linux
/dev/sdd3          411648      616447      102400   83  Linux
/dev/sdd4          616448      923647      153600    5  Extended
/dev/sdd5          618496      923647      152576   83  Linux

Command (m for help): wq
The partition table has been altered!
```

```
Syncing disks.
[root@localhost ~]# mkdir /f1
mkdir: cannot create directory '/f1': File exists
[root@localhost ~]# mkdir /d1
[root@localhost ~]# mkdir /d2
[root@localhost ~]# mkdir /d3
[root@localhost ~]# mkdir /d4
[root@localhost ~]# mkdir /d5
[root@localhost ~]# mkfs.ext4 /dev/sdd1
mke2fs 1.42.9 (28-Dec-2013)
Filesystem label=
OS type: Linux
Block size=1024 (log=0)
Fragment size=1024 (log=0)
Stride=0 blocks, Stripe width=0 blocks
25688 inodes, 102400 blocks
5120 blocks (5.00%) reserved for the super user
First data block=1
Maximum filesystem blocks=33685504
13 block groups
8192 blocks per group, 8192 fragments per group
1976 inodes per group
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]# mkfs.ext4 /dev/sdd2
mke2fs 1.42.9 (28-Dec-2013)
Filesystem label=
OS type: Linux
Block size=1024 (log=0)
Fragment size=1024 (log=0)
Stride=0 blocks, Stripe width=0 blocks
25688 inodes, 102400 blocks
5120 blocks (5.00%) reserved for the super user
First data block=1
Maximum filesystem blocks=33685504
13 block groups
8192 blocks per group, 8192 fragments per group
1976 inodes per group
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]# mkfs.ext4 /dev/sdd3
mke2fs 1.42.9 (28-Dec-2013)
Filesystem label=
OS type: Linux
Block size=1024 (log=0)
Fragment size=1024 (log=0)
Stride=0 blocks, Stripe width=0 blocks
25688 inodes, 102400 blocks
5120 blocks (5.00%) reserved for the super user
```

```
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]# mkfs.ext4 /dev/sdd4
mke2fs 1.42.9 (28-Dec-2013)
mkfs.ext4: inode_size (128) * inodes_count (0) too big for a
        filesystem with 0 blocks, specify higher inode_ratio (-i)
        or lower inode count (-N).

[root@localhost ~]# mkfs.ext4 /dev/sdd5
mke2fs 1.42.9 (28-Dec-2013)
Filesystem label=
OS type: Linux
Block size=1024 (log=0)
Fragment size=1024 (log=0)
Stride=0 blocks, Stripe width=0 blocks
38152 inodes, 152576 blocks
7628 blocks (5.00%) reserved for the super user
First data block=1
Maximum filesystem blocks=33816576
19 block groups
8192 blocks per group, 8192 fragments per group
2008 inodes per group
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]# blkid
/dev/mapper/rhel-var: UUID="978e27f0-7d57-4dc0-bb2d-a1d9346983bf" TYPE="xfs"
/dev/sda2: UUID="tP2dJZ-nC0k-UUg2-BwFA-gPfP-nnv6-0Cli1h" TYPE="LVM2_member"
/dev/sda1: UUID="ad4764ff-05af-4f49-ba8a-507a984872cb" TYPE="xfs"
/dev/sdb2: UUID="72921ac2-7642-4a64-8354-6907083f54f2" TYPE="xfs"
/dev/sdc: UUID="24b39253-0801-427c-b9fc-519651afc93e" TYPE="ext4"
/dev/sdd1: UUID="df4fc180-f77d-4902-b556-d3227471427c" TYPE="ext4"
/dev/sdd2: UUID="97bf69ef-8468-4add-a44d-aa7607f205db" TYPE="ext4"
/dev/sdd3: UUID="67257403-48c5-4d78-9e1a-387139b9e809" TYPE="ext4"
/dev/sr0: UUID="2018-03-22-19-04-59-00" LABEL="RHEL-7.5 Server.x86_64" TYPE="iso9660" PTTYPE="dos"
/dev/mapper/rhel-root: UUID="d64ec991-efea-4da4-b418-aa0a07d3eaaf" TYPE="xfs"
/dev/mapper/rhel-swap: UUID="bbd1e301-23c2-4878-ba3c-f5c14fee2401" TYPE="swap"
/dev/mapper/rhel-home: UUID="a6d58447-ee81-40fc-89a4-a83bb3dc5f35" TYPE="xfs"
/dev/sdd5: UUID="f1af5388-d6eb-476a-a70c-96956f011ba7" TYPE="ext4"
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# mount /d1
[root@localhost ~]# mount /d2
[root@localhost ~]# mount /d3
[root@localhost ~]# mount /d5
[root@localhost ~]# lsblk |grep -i sdd
sdd             8:48   0    3G  0 disk
├─sdd1          8:49   0  100M  0 part /d1
├─sdd2          8:50   0  100M  0 part /d2
├─sdd3          8:51   0  100M  0 part /d3
├─sdd4          8:52   0    1K  0 part
└─sdd5          8:53   0  149M  0 part /d5
[root@localhost ~]#
```

7. Extend a partition and resize the filesystem.

```
[root@localhost ~]# lsblk
NAME           MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda              8:0    0   20G  0 disk
├─sda1           8:1    0  400M  0 part /boot
└─sda2           8:2    0 19.6G  0 part
  ├─rhel-root 253:0    0 10.3G  0 lvm  /
  ├─rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├─rhel-home 253:2    0  956M  0 lvm  /home
  └─rhel-var  253:3    0  6.5G  0 lvm  /var
sdb              8:16   0    2G  0 disk
└─sdb1           8:17   0  500M  0 part
sdc              8:32   0    1G  0 disk /sathii
sdd              8:48   0    3G  0 disk
├─sdd1           8:49   0  100M  0 part /d1
├─sdd2           8:50   0  100M  0 part /d2
├─sdd3           8:51   0  100M  0 part /d3
├─sdd4           8:52   0    1K  0 part
└─sdd5           8:53   0  149M  0 part /d5
sde              8:64   0    1G  0 disk
├─sde1           8:65   0  100M  0 part /e1
└─sde2           8:66   0  200M  0 part /e2
sr0             11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost ~]# umount /e2
[root@localhost ~]# fdisk /dev/sde
Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): d
Partition number (1,2, default 2):
Partition 2 is deleted

Command (m for help): p

Disk /dev/sde: 1073 MB, 1073741824 bytes, 2097152 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0xf303a919

   Device Boot      Start         End      Blocks   Id  System
/dev/sde1            2048      206847      102400   83  Linux

Command (m for help): n
Partition type:
   p   primary (1 primary, 0 extended, 3 free)
   e   extended
Select (default p):
Using default response p
Partition number (2-4, default 2):
First sector (206848-2097151, default 206848):
Using default value 206848
Last sector, +sectors or +size{K,M,G} (206848-2097151, default 2097151): +300M
Partition 2 of type Linux and of size 300 MiB is set

Command (m for help): wq
The partition table has been altered!
```

```
[root@localhost ~]# partprobe
Warning: Unable to open /dev/sr0 read-write (Read-only file system).  /dev/sr0 has been opened read-only.
[root@localhost ~]# df -hT
Filesystem           Type      Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root xfs        11G  3.3G  7.1G  32% /
devtmpfs             devtmpfs  895M     0  895M   0% /dev
tmpfs                tmpfs     911M     0  911M   0% /dev/shm
tmpfs                tmpfs     911M   11M  901M   2% /run
tmpfs                tmpfs     911M     0  911M   0% /sys/fs/cgroup
/dev/sdc             ext4      976M  2.5M  907M   1% /sathii
/dev/sda1            xfs       397M  158M  240M  40% /boot
/dev/mapper/rhel-var  xfs      6.6G  154M  6.4G   3% /var
/dev/mapper/rhel-home xfs      953M   39M  914M   5% /home
tmpfs                tmpfs     183M  4.0K  183M   1% /run/user/42
tmpfs                tmpfs     183M   40K  183M   1% /run/user/1000
/dev/sr0             iso9660   4.4G  4.4G     0 100% /run/media/sathish/RHEL-7.5 Server.x86_64
tmpfs                tmpfs     183M     0  183M   0% /run/user/0
/dev/sdd1            ext4       93M  1.6M   85M   2% /d1
/dev/sdd2            ext4       93M  1.6M   85M   2% /d2
/dev/sdd3            ext4       93M  1.6M   85M   2% /d3
/dev/sdd5            ext4      141M  1.6M  129M   2% /d5
/dev/sde1            ext4       93M  1.6M   85M   2% /e1
[root@localhost ~]# e2fsck -f /dev/sde2
e2fsck 1.42.9 (28-Dec-2013)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/sde2: 11/51200 files (0.0% non-contiguous), 12115/204800 blocks
[root@localhost ~]# resize2fs /dev/sde2
resize2fs 1.42.9 (28-Dec-2013)
Resizing the filesystem on /dev/sde2 to 307200 (1k) blocks.
The filesystem on /dev/sde2 is now 307200 blocks long.

[root@localhost ~]# mount /e2
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda             8:0    0   20G  0 disk
├─sda1          8:1    0  400M  0 part /boot
└─sda2          8:2    0 19.6G  0 part
  ├─rhel-root 253:0    0 10.3G  0 lvm  /
  ├─rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├─rhel-home 253:2    0  956M  0 lvm  /home
  └─rhel-var  253:3    0  6.5G  0 lvm  /var
sdb             8:16   0    2G  0 disk
└─sdb1          8:17   0  500M  0 part
sdc             8:32   0    1G  0 disk /sathii
sdd             8:48   0    3G  0 disk
├─sdd1          8:49   0  100M  0 part /d1
├─sdd2          8:50   0  100M  0 part /d2
├─sdd3          8:51   0  100M  0 part /d3
├─sdd4          8:52   0  512B  0 part
└─sdd5          8:53   0  149M  0 part /d5
sde             8:64   0    1G  0 disk
├─sde1          8:65   0  100M  0 part /e1
└─sde2          8:66   0  300M  0 part /e2
sr0            11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
```

8. Unmount a partition and remove the entry from /etc/fstab.

```
[root@localhost ~]# lsblk
NAME           MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda              8:0    0   20G  0 disk
├─sda1           8:1    0  400M  0 part /boot
└─sda2           8:2    0 19.6G  0 part
  ├─rhel-root 253:0    0 10.3G  0 lvm  /
  ├─rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├─rhel-home 253:2    0  956M  0 lvm  /home
  └─rhel-var  253:3    0  6.5G  0 lvm  /var
sdb              8:16   0    2G  0 disk
└─sdb1           8:17   0  500M  0 part
sdc              8:32   0    1G  0 disk /sathii
sdd              8:48   0    3G  0 disk
├─sdd1           8:49   0  100M  0 part /d1
├─sdd2           8:50   0  100M  0 part /d2
├─sdd3           8:51   0  100M  0 part /d3
├─sdd4           8:52   0  512B  0 part
└─sdd5           8:53   0  149M  0 part /d5
sde              8:64   0    1G  0 disk
├─sde1           8:65   0  100M  0 part /e1
└─sde2           8:66   0  300M  0 part /e2
sr0             11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost ~]# umount /e2
[root@localhost ~]# lsblk
NAME           MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda              8:0    0   20G  0 disk
├─sda1           8:1    0  400M  0 part /boot
└─sda2           8:2    0 19.6G  0 part
  ├─rhel-root 253:0    0 10.3G  0 lvm  /
  ├─rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├─rhel-home 253:2    0  956M  0 lvm  /home
  └─rhel-var  253:3    0  6.5G  0 lvm  /var
sdb              8:16   0    2G  0 disk
└─sdb1           8:17   0  500M  0 part
sdc              8:32   0    1G  0 disk /sathii
sdd              8:48   0    3G  0 disk
├─sdd1           8:49   0  100M  0 part /d1
├─sdd2           8:50   0  100M  0 part /d2
├─sdd3           8:51   0  100M  0 part /d3
├─sdd4           8:52   0  512B  0 part
└─sdd5           8:53   0  149M  0 part /d5
sde              8:64   0    1G  0 disk
├─sde1           8:65   0  100M  0 part /e1
└─sde2           8:66   0  300M  0 part
sr0             11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost ~]# vi /etc/fstab
```

```
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# lsblk
NAME           MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda              8:0    0   20G  0 disk
├─sda1           8:1    0  400M  0 part /boot
└─sda2           8:2    0 19.6G  0 part
  ├─rhel-root 253:0    0 10.3G  0 lvm  /
  ├─rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├─rhel-home 253:2    0  956M  0 lvm  /home
  └─rhel-var  253:3    0  6.5G  0 lvm  /var
sdb              8:16   0    2G  0 disk
└─sdb1           8:17   0  500M  0 part
sdc              8:32   0    1G  0 disk /sathii
sdd              8:48   0    3G  0 disk
├─sdd1           8:49   0  100M  0 part /d1
├─sdd2           8:50   0  100M  0 part /d2
├─sdd3           8:51   0  100M  0 part /d3
├─sdd4           8:52   0  512B  0 part
└─sdd5           8:53   0  149M  0 part /d5
sde              8:64   0    1G  0 disk
├─sde1           8:65   0  100M  0 part /e1
└─sde2           8:66   0  300M  0 part
sr0             11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost ~]#
```

9. Format and mount a disk directly without creating a partition table.

```
└sda2          8:2    0 19.6G  0 part
  ├rhel-root 253:0    0 10.3G  0 lvm  /
  ├rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├rhel-home 253:2    0  956M  0 lvm  /home
  └rhel-var  253:3    0  6.5G  0 lvm  /var
sdb            8:16   0    2G  0 disk
└sdb1          8:17   0  500M  0 part
sdc            8:32   0    1G  0 disk /sathii
sdd            8:48   0    3G  0 disk
├sdd1          8:49   0  100M  0 part /d1
├sdd2          8:50   0  100M  0 part /d2
├sdd3          8:51   0  100M  0 part /d3
├sdd4          8:52   0  512B  0 part
└sdd5          8:53   0  149M  0 part /d5
sde            8:64   0    1G  0 disk
├sde1          8:65   0  100M  0 part /e1
└sde2          8:66   0  300M  0 part
sdf            8:80   0    1G  0 disk
sr0           11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost sathish]# mkdir /f1
mkdir: cannot create directory '/f1': File exists
[root@localhost sathish]# mkdir /df1
[root@localhost sathish]# mkfs.ext4 /dev/sdf
mke2fs 1.42.9 (28-Dec-2013)
/dev/sdf is entire device, not just one partition!
Proceed anyway? (y,n) y
Filesystem label=
OS type: Linux
Block size=4096 (log=2)
Fragment size=4096 (log=2)
Stride=0 blocks, Stripe width=0 blocks
65536 inodes, 262144 blocks
13107 blocks (5.00%) reserved for the super user
First data block=0
Maximum filesystem blocks=268435456
8 block groups
32768 blocks per group, 32768 fragments per group
8192 inodes per group
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost sathish]# blkid
/dev/mapper/rhel-var: UUID="978e27f0-7d57-4dc0-bb2d-a1d9346983bf" TYPE="xfs"
/dev/sda2: UUID="tP2dJZ-nC0k-UUg2-BwFA-gPfP-nnv6-0Cli1h" TYPE="LVM2_member"
/dev/sda1: UUID="ad4764ff-05af-4f49-ba8a-507a984872cb" TYPE="xfs"
/dev/sdc: UUID="24b39253-0801-427c-b9fc-519651afc93e" TYPE="ext4"
/dev/sdd1: UUID="df4fc180-f77d-4902-b556-d3227471427c" TYPE="ext4"
/dev/sdd2: UUID="97bf69ef-8468-4add-a44d-aa7607f205db" TYPE="ext4"
/dev/sdd3: UUID="67257403-48c5-4d78-9e1a-387139b9e809" TYPE="ext4"
/dev/sr0: UUID="2018-03-22-19-04-59-00" LABEL="RHEL-7.5 Server.x86_64" TYPE="iso9660" PTTYPE="dos"
/dev/mapper/rhel-root: UUID="d64ec991-efea-4da4-b418-aa0a07d3eaaf" TYPE="xfs"
/dev/mapper/rhel-swap: UUID="bbd1e301-23c2-4878-ba3c-f5c14fee2401" TYPE="swap"
/dev/mapper/rhel-home: UUID="a6d58447-ee81-40fc-89a4-a83bb3dc5f35" TYPE="xfs"
/dev/sdd5: UUID="f1af5388-d6eb-476a-a70c-96956f011ba7" TYPE="ext4"
/dev/sdb1: UUID="75ea49fd-0311-41b0-93d7-2ee41ef4bd32" TYPE="ext4"
/dev/sde1: UUID="fb2e22c9-e305-409b-9d02-8b35eaa9b7ac" TYPE="ext4"
/dev/sde2: UUID="cd1dee1e-8353-4628-b464-d15536898010" TYPE="ext4"
/dev/sdf: UUID="459abf43-8a2c-41aa-97e4-5582f0bee32b" TYPE="ext4"
[root@localhost sathish]#
```

```
#

# /etc/fstab
# Created by anaconda on Sat May 10 19:27:48 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk'
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info
#
/dev/mapper/rhel-root   /                       xfs     defaults        0 0
UUID=ad4764ff-05af-4f49-ba8a-507a984872cb /boot                   xfs     defaults        0 0
/dev/mapper/rhel-home   /home                   xfs     defaults        0 0
/dev/mapper/rhel-var    /var                    xfs     defaults        0 0
/dev/mapper/rhel-swap   swap                    swap    defaults        0 00
UUID=72921ac2-7642-4a64-8354-6907083f54f2 /Nihal ext4  defaults 0 0
UUID=24b39253-0801-427c-b9fc-519651afc93e /sathii  ext4 defaults 0 0
UUID=df4fc180-f77d-4902-b556-d3227471427c /d1 ext4 defaults 0 0
UUID=97bf69ef-8468-4add-a44d-aa7607f205db /d2 ext4 defaults 0 0
UUID=67257403-48c5-4d78-9e1a-387139b9e809 /d3 ext4 defaults 0 0
UUID=f1af5388-d6eb-476a-a70c-96956f011ba7 /d5 ext4 defaults 0 0
UUID=fb2e22c9-e305-409b-9d02-8b35eaa9b7ac /e1 ext4 defaults 0 0
UUID=459abf43-8a2c-41aa-97e4-5582f0bee32b /df1 ext4 defaults 0 0
```

```
[root@localhost sathish]# vi /etc/fstab
[root@localhost sathish]# mount /df1
[root@localhost sathish]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda             8:0    0   20G  0 disk
├sda1           8:1    0  400M  0 part /boot
└sda2           8:2    0 19.6G  0 part
  ├rhel-root 253:0    0 10.3G  0 lvm  /
  ├rhel-swap 253:1    0  1.9G  0 lvm  [SWAP]
  ├rhel-home 253:2    0  956M  0 lvm  /home
  └rhel-var  253:3    0  6.5G  0 lvm  /var
sdb             8:16   0    2G  0 disk
└sdb1           8:17   0  500M  0 part
sdc             8:32   0    1G  0 disk /sathii
sdd             8:48   0    3G  0 disk
├sdd1           8:49   0  100M  0 part /d1
├sdd2           8:50   0  100M  0 part /d2
├sdd3           8:51   0  100M  0 part /d3
├sdd4           8:52   0  512B  0 part
└sdd5           8:53   0  149M  0 part /d5
sde             8:64   0    1G  0 disk
├sde1           8:65   0  100M  0 part /e1
└sde2           8:66   0  300M  0 part
sdf             8:80   0    1G  0 disk /df1
sr0            11:0    1  4.3G  0 rom  /run/media/sathish/RHEL-7.5 Server.x86_64
[root@localhost sathish]#
```

10. Create both a hard and a soft link to a file and verify with ls -li. SOFTLINK

```
[root@localhost sathish]# cd /e1
[root@localhost e1]# ll
total 12
drwx------. 2 root root 12288 May 17 17:05 lost+found
[root@localhost e1]# touch ASK
[root@localhost e1]# ll
total 13
-rw-r--r--. 1 root root     0 May 17 19:18 ASK
drwx------. 2 root root 12288 May 17 17:05 lost+found
[root@localhost e1]# ln -s /e1/ASK /home
[root@localhost e1]# cd /home
[root@localhost home]# ll
total 4
lrwxrwxrwx.  1 root    root       7 May 17 19:19 ASK -> /e1/ASK
drwx------.  3 gani    gani      78 May 16 21:17 gani
drwx------.  3 kushi   kushi     78 May 13 16:05 kushi
drwx------.  3    1008 gani      78 May 13 21:54 mahi
drwx------.  5 navi    navi     157 May 15 10:46 navi
drwx------.  3 Pawan   Pawan     78 May 12 09:09 Pawan
-rw-r--r--.  1 root    root       0 May 17 11:36 sathii
drwx------. 15 sathish sathish 4096 May 17 15:33 sathish
drwx------.  5 Shiva   Shiva    128 May 16 17:53 Shiva
[root@localhost home]#
```

HARDLINK

```
[root@localhost home]# touch hard
[root@localhost home]# ll
total 4
lrwxrwxrwx.  1 root    root       7 May 17 19:19 ASK -> /e1/ASK
drwx------.  3 gani    gani      78 May 16 21:17 gani
-rw-r--r--.  1 root    root       0 May 17 19:30 hard
drwx------.  3 kushi   kushi     78 May 13 16:05 kushi
drwx------.  3    1008 gani      78 May 13 21:54 mahi
drwx------.  5 navi    navi     157 May 15 10:46 navi
drwx------.  3 Pawan   Pawan     78 May 12 09:09 Pawan
-rw-r--r--.  1 root    root       0 May 17 11:36 sathii
drwx------. 15 sathish sathish 4096 May 17 15:33 sathish
drwx------.  5 Shiva   Shiva    128 May 16 17:53 Shiva
[root@localhost home]# ln hard linkhard
[root@localhost home]# ll
total 4
lrwxrwxrwx.  1 root    root       7 May 17 19:19 ASK -> /e1/ASK
drwx------.  3 gani    gani      78 May 16 21:17 gani
-rw-r--r--.  2 root    root       0 May 17 19:30 hard
drwx------.  3 kushi   kushi     78 May 13 16:05 kushi
-rw-r--r--.  2 root    root       0 May 17 19:30 linkhard
drwx------.  3    1008 gani      78 May 13 21:54 mahi
drwx------.  5 navi    navi     157 May 15 10:46 navi
drwx------.  3 Pawan   Pawan     78 May 12 09:09 Pawan
-rw-r--r--.  1 root    root       0 May 17 11:36 sathii
drwx------. 15 sathish sathish 4096 May 17 15:33 sathish
drwx------.  5 Shiva   Shiva    128 May 16 17:53 Shiva
[root@localhost home]#
```

Ls -li

```
[root@localhost home]# ls -li
total 4
1776 lrwxrwxrwx.  1 root    root       7 May 17 19:19 ASK -> /e1/ASK
1769 drwx------.  3 gani    gani      78 May 16 21:17 gani
1777 -rw-r--r--.  2 root    root       0 May 17 19:30 hard
 120 drwx------.  3 kushi   kushi     78 May 13 16:05 kushi
1777 -rw-r--r--.  2 root    root       0 May 17 19:30 linkhard
 124 drwx------.  3    1008 gani      78 May 13 21:54 mahi
1760 drwx------.  5 navi    navi     157 May 15 10:46 navi
 116 drwx------.  3 Pawan   Pawan     78 May 12 09:09 Pawan
1762 -rw-r--r--.  1 root    root       0 May 17 11:36 sathii
  67 drwx------. 15 sathish sathish 4096 May 17 15:33 sathish
 104 drwx------.  5 Shiva   Shiva    128 May 16 17:53 Shiva
[root@localhost home]#
```

11. Create a new user, assign a password, and verify entries in /etc/passwd, /etc/shadow, and /etc/group.

CREATING user

```
[root@localhost ~]# useradd  -u 1030 -g jagan -c "sathi" -d /var/karthi -s /bin/sh karthi
[root@localhost ~]# passwd karthi
Changing password for user karthi.
New password:
BAD PASSWORD: The password is shorter than 7 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]#
```

/etc/passwd

```
[root@localhost ~]# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:99:99:Nobody:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
polkitd:x:999:998:User for polkitd:/:/sbin/nologin
libstoragemgmt:x:998:996:daemon account for libstoragemgmt:/var/run/lsm:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
colord:x:997:995:User for colord:/var/lib/colord:/sbin/nologin
saslauth:x:996:76:Saslauthd user:/run/saslauthd:/sbin/nologin
abrt:x:173:173::/etc/abrt:/sbin/nologin
setroubleshoot:x:995:992::/var/lib/setroubleshoot:/sbin/nologin
rtkit:x:172:172:RealtimeKit:/proc:/sbin/nologin
gluster:x:994:991:GlusterFS daemons:/var/run/gluster:/sbin/nologin
chrony:x:993:990::/var/lib/chrony:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
nfsnobody:x:65534:65534:Anonymous NFS User:/var/lib/nfs:/sbin/nologin
unbound:x:992:989:Unbound DNS resolver:/etc/unbound:/sbin/nologin
tss:x:59:59:Account used by the trousers package to sandbox the tcsd daemon:/dev/null:/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/sbin/nologin
geoclue:x:991:987:User for geoclue:/var/lib/geoclue:/sbin/nologin
radvd:x:75:75:radvd user:/:/sbin/nologin
qemu:x:107:107:qemu user:/:/sbin/nologin
pulse:x:171:171:PulseAudio System Daemon:/var/run/pulse:/sbin/nologin
gdm:x:42:42::/var/lib/gdm:/sbin/nologin
gnome-initial-setup:x:990:984::/run/gnome-initial-setup/:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
postfix:x:89:89::/var/spool/postfix:/sbin/nologin
ntp:x:38:38::/etc/ntp:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
sathish:x:1000:1000:sathish:/home/sathish:/bin/bash
Shiva:x:1001:1004::/home/Shiva:/bin/bash
Pawan:x:1002:1005::/home/Pawan:/bin/bash
navi:x:1005:1010::/home/navi:/bin/bash
sandhya:x:1006:1011::/home/sandhya:/bin/bash
kushi:x:1007:1007::/home/kushi:/bin/bash
jagan:x:1009:1010:hello:/var:/bin/bash
gani:x:1010:1012::/home/gani:/bin/bash
Ganesh:x:1011:1013:hii:/var:/bin/sh
sathiroy:x:1015:1000:sathii:/home/kushi:/bin/sh
kiran:x:1016:1000:sathii:/var/kiran:/bin/sh
yogi:x:1019:1009:hello:/var/kiran:/bin/sh
sagar:x:1020:1009:hello:/var/abhii:/bin/sh
rupi:x:1021:1009:hello:/home/mahi:/bin/sh
raki:x:1022:1009:hello:/var/abd:/bin/sh
sanju:x:1023:1009:hello:/var/abd:/bin/sh
chitti:x:1024:1024::/home/chitti:/bin/bash
karthi:x:1030:1009:sathi:/var/karthi:/bin/sh
[root@localhost ~]#
```

/etc/shadow

```
[root@localhost ~]# cat /etc/shadow
root:$6$iZRYBFKNV9g5ABiA$Zy9G4U8/dfVOPk7ncMAVrYm5KUxwtSyx0o7eAAG1BkA5niGkqlYhqY9Z46L7LxLQu.fXxvPBWIKo5RLK/Y8Ai0::0:99999:7:::
bin:*:17492:0:99999:7:::
daemon:*:17492:0:99999:7:::
adm:*:17492:0:99999:7:::
lp:*:17492:0:99999:7:::
sync:*:17492:0:99999:7:::
shutdown:*:17492:0:99999:7:::
halt:*:17492:0:99999:7:::
mail:*:17492:0:99999:7:::
operator:*:17492:0:99999:7:::
games:*:17492:0:99999:7:::
ftp:*:17492:0:99999:7:::
nobody:*:17492:0:99999:7:::
systemd-network:!!:20218::::::
dbus:!!:20218::::::
polkitd:!!:20218::::::
libstoragemgmt:!!:20218::::::
rpc:!!:20218:0:99999:7:::
colord:!!:20218::::::
saslauth:!!:20218::::::
abrt:!!:20218::::::
setroubleshoot:!!:20218::::::
rtkit:!!:20218::::::
gluster:!!:20218::::::
chrony:!!:20218::::::
rpcuser:!!:20218::::::
nfsnobody:!!:20218::::::
unbound:!!:20218::::::
tss:!!:20218::::::
usbmuxd:!!:20218::::::
geoclue:!!:20218::::::
radvd:!!:20218::::::
qemu:!!:20218::::::
pulse:!!:20218::::::
gdm:!!:20218::::::
gnome-initial-setup:!!:20218::::::
sshd:!!:20218::::::
avahi:!!:20218::::::
postfix:!!:20218::::::
ntp:!!:20218::::::
tcpdump:!!:20218::::::
sathish:$6$ppCLJPvxINCZzQPk$J9LRGQG.UH30wo/dhqLKMXpom45JBU4xKO8SuWH4HxwgLs5azU3C2/3Ool/XRpp2JPeD.JzwTYMUv5kHZbrEl.::0:99999:7:::
Shiva:$6$7CkKGnBW$isuWqZCjR3QJ6N6zzx9OruQe6Hcyf9QFNOTOt277xow1MuSEMiHZ.10ta3E/ihy3SqD4kpy/L6QbLyUbBG34f0:20221:0:99999:7:::
Pawan:!!:20220:0:99999:7:::
navi:$6$1PHpqYlZ$RoWPTPf/U.5WxA006Bv/tyhZDAfFzmsRQ5V3iubHvuguwJUPVdupEfVzVzaG98zCcSnFGJkvRBYK9lK9D4vyC.:20224:0:99999:7:::
sandhya:$6$aPH32X6D$n3m9ddMFXQ44YgmRPNKyUzU14cvQ8T3as/60195LH52j2EQeq1zfyUMYuVHq5m01Skkjrqt7nObMz5Q/ibJCe0:20225:0:99999:7:::
kushi:!!:20221:0:99999:7:::
jagan:!!:20221:60:120:7::20260101:
gani:$6$qktPjJGr$ufXQ5q.Sr7wm4dJ06.MriFUais30jmhlhjjPj5F4Mnv7yQfFmVBLvLY2/UJCxkM2E06LBQL/wrUQrJQPxtQH71:20224:0:99999:7:::
Ganesh:$6$IDQaPmTS$IgzasMbdTYB84aPXJSCl7/lfhAcoJIWwPw9j2dSL7PzWqOhVsDEItBj0lWYCa21205P.vMqS7K2iDXTIrrora1:20224:0:99999:7:::
sathiroy:!!:20224:0:99999:7:::
kiran:$6$ZS61bWhK$AlW3lm2theZPT7U3Q8OOS6eLOgS1CzSnuhNdPfoDKxamU.OlgsmMeO4/mNTm7YWDn7c1iJFu7KS7srXILdOPe0:20224:0:99999:7:::
yogi:$6$evqKtA/K$NmfboV1ddO34VseGeE0f6zTNiH3MN/Dtzfra27nN85NFUs4taMli4pj.sNHPIyBSUIgmdSCDGs6v.N1W4amIh1:20225:0:99999:7:::
sagar:!!:20225:0:99999:7:::
rupi:$6$Zjc3OZyw$xNX/zFufiUxj4jJzTHwI6HDdJhdPTUbpA3hJ2A39svbORNBViDretORtwqxHZPgq3Oc9qkUHfCVzJ4z.CFAgb1:20225:0:99999:7:::
raki:!!:20225:0:99999:7:::
sanju:!!:20225:0:99999:7:::
chitti:$6$ojiHjBu/$fQcLHNBWbPCr34Z/7vCmDbD7FSSX0ZnYIt4xyUl/Trc009MVgQeO634SgkSDzXdImXprDhT2xbdOVaohywDYt/:20225:0:99999:7:::
karthi:$6$bWYpjthA$PXaiehkRaayWy/8y1dj/OeQn2TLCXVMmJ408Ub6cK2aDV1u.TaYfN7kSgPlyyTbwPxhCO6jwk5KwB4Q.KSM6m1:20225:0:99999:7:::
[root@localhost ~]#
```

12. Create a group, assign users to it, and verify with id.

```
[root@localhost /]# groupadd fun
[root@localhost /]# gpasswd -M sathish,Shiva fun
[root@localhost /]# getent group fun
fun:x:1014:sathish,Shiva
[root@localhost /]#
```

13. Set an ACL to give user1 read/write access to a file owned by user2.

```
[root@localhost /]# setfacl -m u:Shiva:rwx file
[root@localhost /]# su - Shiva
Last login: Fri May 16 18:06:26 IST 2025 on pts/0
[Shiva@localhost ~]$ exit
logout
[root@localhost /]# ls -ld /file
-rw-rwxr--+ 1 root root 18 May 13 09:48 /file
[root@localhost /]#
```

14. Set the sticky bit on a directory and test file deletion permissions.

```
[root@localhost var]#
[root@localhost var]# touch reddy
[root@localhost var]# ll
total 20
drwxr-xr-x.  2 root   root       6 May 17 19:52 abd
drwxr-xr-x.  2 root   root       6 May 17 19:48 abhii
drwxr-xr-x.  2 root   root      19 May 10 19:36 account
drwxr-xr-x.  2 root   root       6 Dec 14  2017 adm
drwxr-xr-x. 13 root   root     159 May 10 19:48 cache
drwxr-xr-x.  2 root   root       6 Feb 13  2018 crash
drwxr-xr-x.  3 root   root      34 May 10 19:36 db
drwxr-xr-x.  3 root   root      18 May 10 19:36 empty
drwxr-xr-x.  2 root   root       6 Dec 14  2017 games
drwxr-xr-x.  2 root   root       6 Dec 14  2017 gopher
drwx------.  3 karthi jagan     78 May 17 22:41 karthi
drwxr-xr-x.  3 root   root      18 May 10 19:31 kerberos
drwx------.  5 kiran  sathish  142 May 16 21:51 kiran
drwxr-xr-x. 57 root   root    4096 May 10 19:48 lib
drwxr-xr-x.  2 root   root       6 Dec 14  2017 local
lrwxrwxrwx.  1 root   root      11 May 10 19:30 lock -> ../run/lock
drwxr-xr-x. 17 root   root    4096 May 18 10:21 log
lrwxrwxrwx.  1 root   root      10 May 10 19:30 mail -> spool/mail
drwxr-xr-x.  2 root   root       6 Dec 14  2017 nis
drwxr-xr-x.  2 root   root       6 Dec 14  2017 opt
drwxr-xr-x.  2 root   root       6 Dec 14  2017 preserve
-rw-r--r--.  1 root   root       0 May 18 10:27 reddy
lrwxrwxrwx.  1 root   root       6 May 10 19:30 run -> ../run
drwxr-xr-x. 14 root   root     167 May 10 19:36 spool
drwxrwxrwt. 54 root   root    8192 May 18 10:22 tmp
drwxr-xr-x.  2 root   root       6 Dec 14  2017 yp
[root@localhost var]# chmod +t /var/reddy
[root@localhost var]# ll
total 20
drwxr-xr-x.  2 root   root       6 May 17 19:52 abd
drwxr-xr-x.  2 root   root       6 May 17 19:48 abhii
drwxr-xr-x.  2 root   root      19 May 10 19:36 account
drwxr-xr-x.  2 root   root       6 Dec 14  2017 adm
drwxr-xr-x. 13 root   root     159 May 10 19:48 cache
drwxr-xr-x.  2 root   root       6 Feb 13  2018 crash
drwxr-xr-x.  3 root   root      34 May 10 19:36 db
drwxr-xr-x.  3 root   root      18 May 10 19:36 empty
drwxr-xr-x.  2 root   root       6 Dec 14  2017 games
drwxr-xr-x.  2 root   root       6 Dec 14  2017 gopher
drwx------.  3 karthi jagan     78 May 17 22:41 karthi
drwxr-xr-x.  3 root   root      18 May 10 19:31 kerberos
drwx------.  5 kiran  sathish  142 May 16 21:51 kiran
drwxr-xr-x. 57 root   root    4096 May 10 19:48 lib
drwxr-xr-x.  2 root   root       6 Dec 14  2017 local
lrwxrwxrwx.  1 root   root      11 May 10 19:30 lock -> ../run/lock
drwxr-xr-x. 17 root   root    4096 May 18 10:21 log
lrwxrwxrwx.  1 root   root      10 May 10 19:30 mail -> spool/mail
drwxr-xr-x.  2 root   root       6 Dec 14  2017 nis
drwxr-xr-x.  2 root   root       6 Dec 14  2017 opt
drwxr-xr-x.  2 root   root       6 Dec 14  2017 preserve
-rw-r--r-T.  1 root   root       0 May 18 10:27 reddy
lrwxrwxrwx.  1 root   root       6 May 10 19:30 run -> ../run
drwxr-xr-x. 14 root   root     167 May 10 19:36 spool
drwxrwxrwt. 54 root   root    8192 May 18 10:22 tmp
drwxr-xr-x.  2 root   root       6 Dec 14  2017 yp
```

```
[root@localhost var]# ls -ld reddy
-rw-r--r-T. 1 root root 0 May 18 10:27 reddy
[root@localhost var]# su - navi
Last login: Sun May 18 10:29:11 IST 2025 on pts/0
this is bashrc
This is bash_profile
[navi@localhost ~]$ cd /var
[navi@localhost var]$ rm -rf reddy
rm: cannot remove 'reddy': Permission denied
[navi@localhost var]$ ls -ld reddy
-rw-r--r-T. 1 root root 0 May 18 10:27 reddy
[navi@localhost var]$
```

15. Set the setuid bit on /usr/bin/passwd and explain the result.

```
[root@localhost /]# chmod u+s /usr/bin/passwd
[root@localhost /]# su - chitti
Last login: Sat May 17 20:32:50 IST 2025 on pts/1
[chitti@localhost ~]$ passwd
Changing password for user chitti.
Changing password for chitti.
(current) UNIX password:
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[chitti@localhost ~]$
```

```
[root@localhost /]# chmod u-s /usr/bin/passwd
[root@localhost /]# su - chitti
Last login: Sat May 17 20:36:41 IST 2025 on pts/1
[chitti@localhost ~]$ passwd
Changing password for user chitti.
Changing password for chitti.
(current) UNIX password:
New password:
BAD PASSWORD: The password contains the user name in some form
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
New password:
Retype new password:
passwd: Authentication token manipulation error
[chitti@localhost ~]$
```

16. Set the setgid bit on a directory and create a file to observe the group ownership.

```
[root@localhost /]# gpasswd -M navi,jagan gani
[root@localhost /]# getent group gani
gani:x:1012:navi,jagan
[root@localhost /]# ls -ld sathii
drwxrwxrwx+ 2 root root 4096 May 17 15:25 sathii
[root@localhost /]# chmod g=rw /sathii
[root@localhost /]# ls -ld sathii
drwxrw-rwx+ 2 root root 4096 May 17 15:25 sathii
```

```
[root@localhost /]# chmod g+s /sathii
[root@localhost /]# ls -ld /sathii
drwxrwSrwx+ 2 root root 4096 May 17 20:59 /sathii
[root@localhost /]# su - sathish
Last login: Sat May 17 20:57:35 IST 2025 on pts/1
bash: Thish: command not found...
bash: This: command not found...
[sathish@localhost ~]$ cd /sathii
[sathish@localhost sathii]$ touch ad
[sathish@localhost sathii]$ ll
total 0
-rw-rw-r--. 1 sathish root    0 May 17 21:02 ad
-rw-rw-r--. 1 sathish sathish 0 May 17 20:59 as
-rw-r--r--. 1 root    root    0 May 16 18:27 file02
-rw-r--r--. 1 root    root    0 May 14 10:59 test01
[sathish@localhost sathii]$
```

17. Define a variable, export it, and verify it with env.

```
[root@localhost home]# cd navi
[root@localhost navi]# MY_VAR="sathish"
[root@localhost navi]# export MY_VAR
[root@localhost navi]# env |grep MY_VAR
MY_VAR=sathish
[root@localhost navi]#
```

18. Use find to locate files larger than 100MB modified in the last 7 days.

```
[root@localhost navi]# find / -type f -size +100M -mtime -7
/proc/kcore
find: '/proc/19712/task/19712/fdinfo/5': No such file or directory
find: '/proc/19712/fdinfo/6': No such file or directory
find: '/run/user/1000/gvfs': Permission denied
/sys/devices/pci0000:00/0000:00:0f.0/resource1_wc
/sys/devices/pci0000:00/0000:00:0f.0/resource1
[root@localhost navi]#
```

19. Extract the second column from /etc/passwd using cut.

```
[root@localhost navi]# cut -d: -f2 /etc/passwd
x
x
x
```

```
[root@localhost /]# grep '/bin/bash' /etc/passwd
root:x:0:0:root:/root:/bin/bash
sathish:x:1000:1000:sathish:/home/sathish:/bin/bash
Shiva:x:1001:1004::/home/Shiva:/bin/bash
Pawan:x:1002:1005::/home/Pawan:/bin/bash
navi:x:1005:1010::/home/navi:/bin/bash
sandhya:x:1006:1011::/home/sandhya:/bin/bash
kushi:x:1007:1007::/home/kushi:/bin/bash
jagan:x:1009:1010:hello:/var:/bin/bash
gani:x:1010:1012::/home/gani:/bin/bash
chitti:x:1024:1024::/home/chitti:/bin/bash
[root@localhost /]#
```

21. Replace "test" with "prod" in a file using sed.

```
[root@localhost /]# vi abc
[root@localhost /]# cat abc
hello
hii
test
[root@localhost /]# sed -i 's/test/prod/g' abc
[root@localhost /]# cat abc
hello
hii
prod
[root@localhost /]#
```

22. Print specific columns using awk.

```
[root@localhost ~]# vi ramu
[root@localhost ~]# awk '{print $1}' ramu
sathi
ram
sri
gani
vamsi
yogi
[root@localhost ~]# awk '{print $1, $2}' ramu
sathi 01
ram 02
sri 03
gani 04
vamsi 05
yogi 06
[root@localhost ~]# awk '{print $1, $2, $3}' ramu
sathi 01 linux
ram 02 windows
sri 03 apple
gani 04 samsung
vamsi 05 iphone
yogi 06 vivo
[root@localhost ~]#
```